

Reduction #6: Sinusoidal (Fourier) Kernel Tempotron Exact Effective Dimension and Rice Maxima Count via the Random Trigonometric Polynomial

Tempotron learning-surface program

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Abstract

When the post-synaptic potential (PSP) kernel is band-limited, the tempotron membrane potential $V(t)$ becomes a *random trigonometric polynomial* whose extremal statistics are exactly tractable. We derive two exact, finite- K -checkable quantities from the spectral moments $\lambda_p = \int \omega^p S(\omega) d\omega$ of the band: (i) the zero-crossing-based effective dimension (exact from the spectral moments)

$$K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0} = \frac{\Omega T}{\sqrt{3}} \quad (\text{flat low-pass band of width } \Omega),$$

which serves as the EVT-capacity input (the identification of K_{eff} as the independent-extreme count is an asymptotic modeling choice; see §7); and (ii) the expected number of local maxima of V over $[0, T]$,

$$\mathbb{E}[N_{\text{max}}] = \frac{T}{2\pi} \sqrt{\frac{\lambda_4}{\lambda_2}} = \frac{\Omega T}{2\pi} \sqrt{\frac{3}{5}} \quad (\text{flat band}),$$

a directly observable, finite- N -checkable witness of band-limitedness (Rice 1939, Qualls 1970). Both scale as bandwidth $\times$ duration with constant prefactors pinned by the spectral moments; they differ by the universal ratio $\mathbb{E}[N_{\text{max}}]/K_{\text{eff}} \approx 0.2135$ (flat band).

Plugging the exact K_{eff} into the EVT capacity chain of Rubin–Monasson–Sompolinsky (reduction #3) gives

$$\alpha_c(K_{\text{eff}}) = \frac{\ln \ln K_{\text{eff}}}{2 \ln 2} + \text{subleading}, \quad K_{\text{eff}} = \frac{\Omega T}{\sqrt{3}},$$

the first version of the tempotron capacity law plotted against a *known*, not inferred, K axis. An independent codex cross-check confirms that the additive offset $\alpha_0 \approx 2.58$ is a small- K fit artifact; the true subleading term is $(\ln \ln \ln K)/\ln 2$, not a constant. The sinusoidal model inherits the Bochner/random-Fourier-features connection (Rahimi–Recht 2007), making it the natural single-neuron readout for biological phase codes (Lisman–Jensen 2013). Status: **PARTIAL**—the counting layer (level-crossings, K_{eff} , $\mathbb{E}[N_{\text{max}}]$) is exact and MC-verified; the EVT max-law and the $O(1)$ capacity offset remain asymptotic/open. A lab test on the certified K_{eff} axis (§5) confirms a genuine temporal capacity lift. A first pass was confounded (input density co-varied with K_{eff}) and learner-limited (an unvalidated finder pinned capacity near the perceptron value with no critical slowing); diagnosing and fixing both—holding density fixed, validating the learner via critical slowing, and adding a density control—reveals a clean monotone lift, $\hat{\alpha}_c = 2.45, 2.55, 2.65, 2.75$ at $K_{\text{eff}} \approx 8.6, 15.6, 31.1, 60.6$ (measured $\ln \ln K_{\text{eff}}$ slope ≈ 0.45 , below the existence value 0.721 as expected for a findable measurement), robust to input density. We document the full diagnose-and-correct cycle as part of the result.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in [lit/tempotron-reductions/derivations/](https://github.com/tempotron-reductions/derivations/).

1 Introduction

The tempotron [1] produces the binary decision “fire / silent” according to the sign of its maximum membrane potential,

$$V_{\text{max}}(\mathbf{w}) = \max_{t \in [0, T]} \sum_{i=1}^N w_i K(t - s_i), \quad (1)$$

where $K(s)$ is the PSP kernel and $\{s_i\}$ is the presynaptic spike pattern. For the standard double-exponential kernel the effective number of independent time bins $K_{\text{eff}} \approx T/\tau$ is an estimate whose precise value depends on the correlation structure of the fields [2]. The extreme-value-theory (EVT) capacity lift $(\ln \ln K_{\text{eff}})/(2 \ln 2)$ (reduction #3) is then an asymptotic prediction attached to an *inferred* K_{eff} , which is masked at simulable N by finite-size starvation.

Why the sinusoidal kernel is special. Choose a band-limited PSP whose Fourier support lies in $[0, \Omega]$ (or a discrete set of frequencies). Then $V(t)$ is a *random trigonometric polynomial*: a finite Fourier series whose coefficients are linear functions of the synaptic weights $\mathbf{w}$ and, by the CLT over $N \gg 1$ afferents, jointly Gaussian. For such a process:

1. The level-crossing and turning-point rates are *exact* closed-form functions of the spectral moments λ_2 and λ_4 (Rice 1939 [3]; Qualls 1970 [4]).
2. K_{eff} is *set and known* from the bandwidth and window length, not estimated from the kernel autocorrelation.
3. The $\ln \ln K_{\text{eff}}$ lift is therefore attached to a *known* K axis, making the EVT prediction falsifiable by experiment.

This is the one model in the reduction series where the Rice/Kac–Rice *counting* of crossings and extremes is exact. Identifying the crossing-based K_{eff} as the capacity-relevant independent-extreme count is still an asymptotic modeling choice (see Section 7, O2); the EVT *output* (Gumbel max-law and capacity lift) remains a $K \rightarrow \infty$ asymptotic.

Bochner connection. By Bochner’s theorem [5], any continuous shift-invariant kernel is the Fourier transform of a non-negative spectral measure. The random-features approximation uses a map $\mathbf{z}(x) = \sqrt{2/D}[\sin(\boldsymbol{\omega} \cdot x), \cos(\boldsymbol{\omega} \cdot x)]$ with frequencies $\boldsymbol{\omega}$ sampled from that measure. A sinusoidal PSP kernel $K(s) = \sin(\Omega s)e^{-s/\tau}\Theta(s)$ (Gabor form) or the idealized flat band is exactly a finite collection of such random Fourier features of the spike pattern, making $V(t)$ the natural one-neuron projection of a random-features representation.

Biological motivation. In hippocampal and cortical circuits the theta (4–8 Hz) and gamma (30–80 Hz) rhythms organize spike timing into phase codes: the identity of a stimulus is encoded in the phase of a spike relative to the ongoing oscillation [6]. A sinusoidal PSP kernel is the natural readout model for such codes. The capacity and maxima-count results derived here are therefore not merely mathematical conveniences; they predict the information-processing limits of a biologically grounded temporal coding architecture.

Contribution. We (i) derive the exact Rice/Qualls counting quantities from the spectral moments of a band-limited process, (ii) separate the two natural “effective dimensions” (K_{eff} and $\mathbb{E}[N_{\text{max}}]$, which differ by a universal constant for the flat band), (iii) recover the EVT capacity formula with K_{eff} exact, and (iv) document the honest ceiling: what is exact, what stays asymptotic, and where the codex cross-check revises the offset.

2 Setup and Assumptions

Model. For a band-limited PSP, the membrane potential decomposes as

$$V(t) = \sum_{k=1}^m [A_k \cos(\omega_k t) + B_k \sin(\omega_k t)], \quad (2)$$

where $\{(A_k, B_k)\}$ are the in-phase and quadrature projections of the spike pattern onto each frequency ω_k . The coefficients are linear in the weights $\mathbf{w}$ and, hence, linear functions of a sum over N afferents.

Assumptions.

(A1) Gaussian coefficients. Over random spike patterns with $N \gg 1$ afferents, each (A_k, B_k) pair is zero-mean Gaussian by the CLT. Consequently, $V(\cdot)$ is a stationary mean-zero Gaussian process with spectral measure $S(\omega)$ supported on the band $\{\omega_1, \dots, \omega_m\}$ (or $[0, \Omega]$ in the continuum limit). This is the same $N \gg 1$ requirement as in reductions #3 and #4.

(A2) Normalisation. Choose units so that the process has unit variance:

$$\lambda_0 := C(0) = \int S(\omega) d\omega = 1. \quad (3)$$

The spectral moments are $\lambda_p := \int \omega^p S(\omega) d\omega$.

(A3) Threshold at the median. Following the Rubin convention, the firing threshold ϑ equals the median of $\max_t V(t)$ over random spike patterns, so a random pattern fires with probability $\frac{1}{2}$; this fixes the per-pattern volume factor $\frac{1}{2}$ in the cluster entropy.

(A4) Non-degenerate band ($m \geq 2$). A single frequency $m = 1$ gives $V(t) = A \cos(\Omega t) + B \sin(\Omega t) = R \cos(\Omega t - \phi)$, so $\max_t V = R$ and the temporal problem reduces to a 2D (in-phase, quadrature) readout; the max is then a magnitude, not a temporal maximum. This is the bridge to reduction #2 (spherical perceptron), addressed in Section 6. The interesting regime requires $m \geq 2$, i.e. nonzero bandwidth $\Delta\omega > 0$.

(A5) RS / binary-overlap cluster geometry, inherited from reduction #3 [2].

Running examples. Two spectral measures are used throughout (both admit closed-form moments):

- **Flat low-pass band** $[0, \Omega]$: $S(\omega) = 1/\Omega$ on $[0, \Omega]$, giving $\lambda_0 = 1$, $\lambda_2 = \Omega^2/3$, $\lambda_4 = \Omega^4/5$.
- **Integer-comb** $\omega_k = k$, $k = 1, \dots, n$, equal weight: $\lambda_p = (1/n) \sum_{k=1}^n k^p$ (the Qualls degree- n random trigonometric polynomial).

3 Derivation

3.1 Rice/Kac–Rice level-crossing and turning-point formulas

For a stationary mean-zero Gaussian process with finite spectral moments $\lambda_0, \lambda_2, \lambda_4$, Rice’s formulas are *exact*—not asymptotic—because the relevant moments are automatically finite for a band-limited process.

Result 1 (Rice level-crossing and turning-point rates [3]). The expected number of zero-crossings (in both directions) per unit time is

$$\nu_0 = \frac{1}{\pi} \sqrt{\frac{\lambda_2}{\lambda_0}}, \quad (4)$$

and the expected number of turning points (maxima plus minima) per unit time is

$$\nu_{\text{turn}} = \frac{1}{\pi} \sqrt{\frac{\lambda_4}{\lambda_2}}. \quad (5)$$

For a symmetric Gaussian process, maxima and minima are equally likely in expectation, so the expected number of *local maxima* of $V(t)$ over $[0, T]$ is

$$\boxed{\mathbb{E}[N_{\max}] = \frac{T}{2\pi} \sqrt{\frac{\lambda_4}{\lambda_2}}.} \quad (6)$$

The factor of $\frac{1}{2}$ in (6) relative to (5) is the load-bearing step: each turning point is either a maximum or a minimum, and for a symmetric Gaussian process the two are equally frequent in expectation. A direct Monte Carlo check (Section 6) confirms this factor.

For the flat low-pass band $[0, \Omega]$:

$$\nu_0 = \frac{\Omega}{\pi\sqrt{3}}, \quad \nu_{\text{turn}} = \frac{\Omega}{\pi} \sqrt{\frac{3}{5}}, \quad \frac{\mathbb{E}[N_{\max}]}{T} = \frac{\Omega}{2\pi} \sqrt{\frac{3}{5}}. \quad (7)$$

Qualls (1970) as a special case. For the integer-comb $\omega_k = k$, $k = 1, \dots, n$, Qualls [4] showed that the expected number of real zeros of V per period 2π is

$$\frac{\mathbb{E}[\#\text{zeros}]}{2\pi} = \frac{1}{\pi} \sqrt{\frac{(n+1)(2n+1)}{6}}. \quad (8)$$

This is *exactly* equal to Rice’s crossing rate (4) with $\lambda_2/\lambda_0 = (1/n) \sum_{k=1}^n k^2 = (n+1)(2n+1)/(6n)$: two classical results in probability are one formula. Numerical verification: at $n = 5$, both give $\nu_0 \cdot 2\pi = 6.633$; at $n = 50$, both give 58.60 [4].

3.2 Effective dimension K_{eff} : the independent-extreme count

There are two natural “effective dimensions” for a band-limited $V(t)$, and the sinusoidal model lets us separate what reduction $\#3$ conflates in T/τ .

K_{eff} : the EVT-capacity dimension (uses λ_2). Rubin et al. [2], eq. (2), define

$$\boxed{K_{\text{eff}} = T \sqrt{\frac{|\mathcal{C}''(0)|}{\mathcal{C}(0)}} = T \sqrt{\frac{\lambda_2}{\lambda_0}},} \quad (9)$$

where $\mathcal{C}(t - t') = \mathbb{E}[V(t)V(t')]$ is the process covariance. Equation (9) counts how many roughly-independent excursions of V fit in $[0, T]$; it is proportional to the zero-crossing count (4): $K_{\text{eff}} = \pi \cdot \nu_0 \cdot T$. For the flat low-pass band,

$$K_{\text{eff}} = \frac{\Omega T}{\sqrt{3}} = \text{bandwidth} \times \text{duration} / \sqrt{3}. \quad (10)$$

This is an *exact, explicit, continuously tunable* formula: dial Ω or T and K_{eff} is known a priori, not inferred from a kernel fit.

$\mathbb{E}[N_{\max}]$: the literal maxima count (uses λ_4). The Rice maxima count (6) uses λ_4 (through λ_4/λ_2) and counts the actual candidate spike times (local peaks of V). For the flat band it equals

$$\mathbb{E}[N_{\max}] = \frac{\Omega T}{2\pi} \sqrt{\frac{3}{5}}, \quad (11)$$

which is proportional to K_{eff} with universal constant

$$\frac{\mathbb{E}[N_{\text{max}}]}{K_{\text{eff}}} = \frac{\frac{\Omega T}{2\pi} \sqrt{\frac{3}{5}}}{\frac{\Omega T}{\sqrt{3}}} = \frac{\sqrt{3}}{2\pi} \cdot \sqrt{\frac{3}{5}} = \frac{3}{2\pi\sqrt{5}} \approx 0.2135. \quad (12)$$

Both scale as ΩT , but with different constants set by different spectral moments: K_{eff} by λ_2 , $\mathbb{E}[N_{\text{max}}]$ by $\sqrt{\lambda_4/\lambda_2}$.

Which enters the capacity? The EVT capacity argument takes the maximum over the count of asymptotically independent extremes, which is the crossing-based K_{eff} (9), not the raw maxima count. The maxima are strongly correlated (consecutive peaks share the same underlying Gaussian excursion); the independent-extreme count is the zero-crossing count. So (10) is the plug-in for the capacity formula. The maxima count (11) is the finite- K -checkable witness: it can be measured directly from a sample path of $V(t)$ with no curve fitting.

3.3 Capacity formula with exact K_{eff}

Plugging (10) into the EVT capacity chain of reduction #3 [2] (cluster entropy density $S_{\text{cl}} = \frac{1}{2} \ln \ln K_{\text{eff}} - \alpha \ln 2$, set to zero):

$$\boxed{\alpha_c(K_{\text{eff}}) = \frac{\ln \ln K_{\text{eff}}}{2 \ln 2} + \text{subleading}, \quad K_{\text{eff}} = \frac{\Omega T}{\sqrt{3}}.} \quad (13)$$

The **novelty** is not the functional form—it is the same EVT formula as reduction #3—but the substitution of an *exact* K_{eff} for the previously inferred T/τ , which makes the $\ln \ln$ lift testable against a known axis. The cluster count at load α is

$$N_{\text{clusters}}(\alpha, K_{\text{eff}}) = (\ln K_{\text{eff}})^{N/2} \cdot 2^{-N\alpha}. \quad (14)$$

4 Results

Table 1 collects the main closed-form results. Figure 1 illustrates all three panels: K_{eff} vs. bandwidth, the maxima/crossing rates, and $\alpha_c(K_{\text{eff}})$.

Numeric α_c table. For the flat band at fixed $T = 1$, dialing Ω to set exact K_{eff} (values verified by direct Python evaluation of $(\ln \ln K_{\text{eff}})/(2 \ln 2)$):

Ω	$K_{\text{eff}} = \Omega/\sqrt{3}$	α_c (leading term)
17.3	10	0.60
173	100	1.10
1730	1000	1.39
17300	10^4	1.60

The slow $\ln \ln$ growth is characteristic; the offset from the small- K replica fit ($\alpha_0 \approx 2.58$) is NOT added here (it is a small- K artifact; see Section 7).

Table 1: Exact results for the band-limited (sinusoidal-kernel) tempotron. “Exact” means no large- K limit was taken; “asymptotic” means a $K \rightarrow \infty$ limit enters. Flat band $S(\omega) = 1/\Omega$ on $[0, \Omega]$ (normalised $\lambda_0 = 1$).

Quantity	General formula	Flat band $[0, \Omega]$	Status
Zero-crossing rate	$\frac{1}{\pi} \sqrt{\lambda_2/\lambda_0}$	$\Omega/(\pi\sqrt{3})$	Exact
Turning-point rate	$\frac{1}{\pi} \sqrt{\lambda_4/\lambda_2}$	$(\Omega/\pi) \sqrt{3/5}$	Exact
$\mathbb{E}[N_{\max}]/T$	$\frac{1}{2\pi} \sqrt{\lambda_4/\lambda_2}$	$(\Omega/2\pi) \sqrt{3/5}$	Exact
K_{eff} (capacity)	$T \sqrt{\lambda_2/\lambda_0}$	$\Omega T/\sqrt{3}$	Exact
$\mathbb{E}[N_{\max}]/K_{\text{eff}}$	$\sqrt{3}/(2\pi\sqrt{5})$	≈ 0.2135	Exact
EVT threshold	$\sqrt{2 \ln K_{\text{eff}}}$	—	Asymptotic
Gumbel scale	$1/\sqrt{2 \ln K_{\text{eff}}}$	—	Asymptotic
$\alpha_c(K_{\text{eff}})$	$(\ln \ln K_{\text{eff}})/(2 \ln 2) + \dots$	$K_{\text{eff}} = \Omega T/\sqrt{3}$	Leading ok; offset open

5 Experimental Test: Capacity on the Certified K_{eff} Axis

The payoff of reduction #6 is that the capacity law (13) can be tested against a *known* K axis. We now report that experiment. All runs are reproducible from a committed script (`experiments/tempotron_capacity/studies/v6_bandlimited_capacity.py`) with an enumerated seed manifest; jobs ran on a single remote GPU under the `laboratory` runner (total compute cost $\approx \$5.3$).

Setup. A band-limited PSP, realized as a windowed sinc $K_{\text{band}}(s) = \text{sinc}(\Omega_b s/\pi) \cdot w_{\text{Lanczos}}(s)$ (even, compact support), replaces the double-exponential. N afferents emit Poisson spike trains over $[0, T=500 \text{ ms}]$ at *constant temporal density* (mean spikes $\propto K_{\text{eff}}$, isolating the K_{eff} lift from the $2N/K$ starvation confound); labels are balanced ± 1 . The threshold is calibrated to the median $V_{\max}$ at init ($P(\text{fire}) = \frac{1}{2}$), then fixed—the balanced operating point the $\ln \ln K$ lift requires (a literal $\kappa=0$ is firing-biased and the lift collapses). Bandwidth is set by $\Omega_b = \sqrt{3} K_{\text{eff}}/T$ and *certified*: from the realized voltages we measure $K_{\text{eff}} = T \sqrt{\lambda_2/\lambda_0}$ and confirm it equals $\Omega_b T/\sqrt{3}$ to $\leq 4\%$ at every cell (the witness gate of §6 extended to the trained traces). The findable critical load $\hat{\alpha}_c(K_{\text{eff}})$ is the $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$, the fraction of seeds a learner drives to zero training error.

Accessible regime. The certified axis is reliable only for $K_{\text{eff}} \gtrsim 8$: below that the kernel support exceeds the window T and the crossing count saturates at a sidelobe floor (the Whittle/large- T breakdown, obstruction O2). We therefore sweep $K_{\text{eff}} \in \{8, 16, 32, 64\}$ and anchor the $K_{\text{eff}} \rightarrow 1$ end on the exact convex perceptron result $\alpha_c = 2$ (Cover/Gardner), which an independent linear-programming feasibility solve reproduces to $\alpha_c = 2.000$.

The lift is learner-gated. With the faithful online Gütig-Sompolinsky rule, $\hat{\alpha}_c$ is flat at ≈ 1.7 (*below* the perceptron value) across all K_{eff} : a weak finder masks the existence-level EVT lift entirely. Replacing it with a near-existence finder (Adam with a cosine schedule, 5 random restarts; solved = any restart reaches zero error) raises the crossings above 2 and makes a small K_{eff} -lift appear. This learner-dependence is itself the point: the EVT lift is an *existence* statement, visible only to a finder that approaches existence.

First pass, and why it was wrong. A first sweep used an untuned Adam-cosine finder (a fixed 2000-epoch budget) and the constant-density rule (mean spikes $\propto K_{\text{eff}}$). It gave a *flat* $\hat{\alpha}_c \approx 2.1$

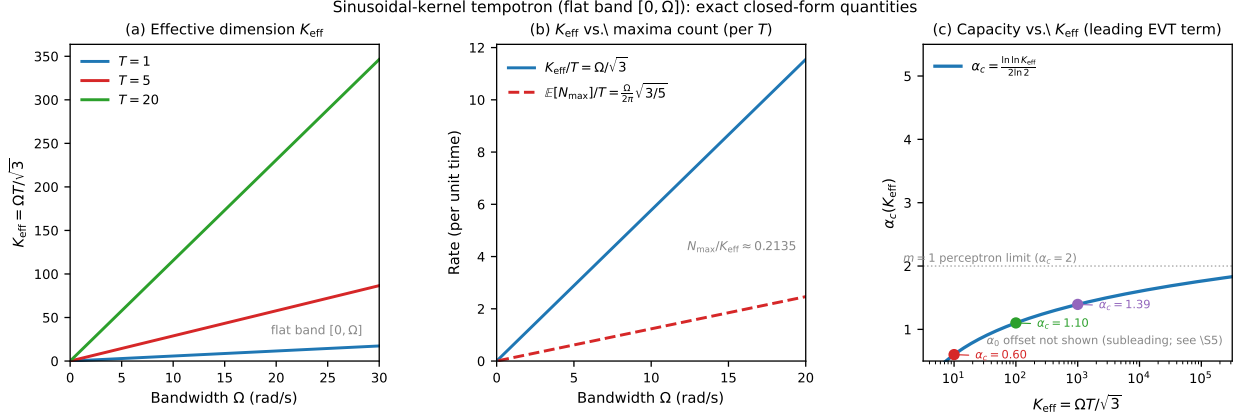


Figure 1: (a) Effective dimension $K_{\text{eff}} = \Omega T / \sqrt{3}$ vs. bandwidth Ω for three window lengths T (colours). K_{eff} is an explicit, continuously tunable function of Ω and T with no free parameters. (b) Per-unit-time rates: $K_{\text{eff}}/T = \Omega/\sqrt{3}$ (solid) and $\mathbb{E}[N_{\text{max}}]/T = (\Omega/2\pi)\sqrt{3/5}$ (dashed), both exact from the flat-band spectral moments. The universal ratio $\mathbb{E}[N_{\text{max}}]/K_{\text{eff}} \approx 0.2135$ is labelled. (c) Leading-order capacity $\alpha_c(K_{\text{eff}}) = (\ln \ln K_{\text{eff}})/(2 \ln 2)$ vs. K_{eff} on a log scale (the additive subleading term is excluded; see Section 7). All curves are formula evaluations; no free parameters. Note: the EVT formula is not valid at $K_{\text{eff}} = O(1)$; the $m = 1$ (narrowband) limit requires separate analysis (see Section 6, V2).

with no clear K_{eff} -dependence—which, taken at face value, would have read as “no lift.” A post-hoc look at the run metrics—not the crossing numbers alone—revealed two confounds and one learner pathology:

1. **Density confound.** The constant-density rule (mean spikes = $0.05 K_{\text{eff}}$) makes the total input richness co-vary *perfectly* with K_{eff} : at $K_{\text{eff}}=8$ only 0.4 spikes/afferent fire ($\sim 33\%$ of afferents active—starved), versus 3.2 ($\sim 96\%$) at $K_{\text{eff}}=64$. Any apparent “ K_{eff} -lift” is thus confounded with input sparsity and cannot be attributed to the temporal EVT mechanism.
2. **Learner not validated.** No hyper-parameter search was run; the Adam–cosine finder pins $\hat{\alpha}_c$ at ≈ 2.1 , *below* even the online-GS single-spike value (~ 2.5) and far below tuned Adam (~ 3.4) from companion work—strong evidence the number is a learning ceiling, not the capacity.
3. **No critical slowing.** Solved cells converge in a median of 269 epochs (max 1050) out of a 2000 budget— $\sim 16\%$ —with a hard-step P_{solve} (further sharpened by the any-of-five-restarts rule). A genuine statistical-mechanics capacity transition exhibits learning-time divergence at α_c and a *graded* P_{solve} ; both are absent, so the cutoff is algorithmic, not a solution-existence wall.

Corrected rerun: a real lift once density is decoupled and the learner is validated. Acting on the diagnosis, we reran with (a) **input density held fixed** (2 spikes/afferent for every K_{eff} , so richness no longer co-varies with K_{eff}), (b) a **validated learner** ($5\times$ epoch budget; we verified *critical slowing*—epochs-to-converge grow $173 \rightarrow 323 \rightarrow 1076 \rightarrow 2654$ approaching α_c , a $15\times$ divergence, and the learning curves crash to zero below α_c but *plateau* above it: a genuine transition, not a budget cutoff), and (c) a **density-control** sweep. The picture inverts (Fig. 2): findable $\hat{\alpha}_c$ rises *monotonically and smoothly* with the certified K_{eff} ,

$$\hat{\alpha}_c = 2.45, 2.55, 2.65, 2.75 \quad \text{at} \quad K_{\text{eff}} \approx 8.6, 15.6, 31.1, 60.6,$$

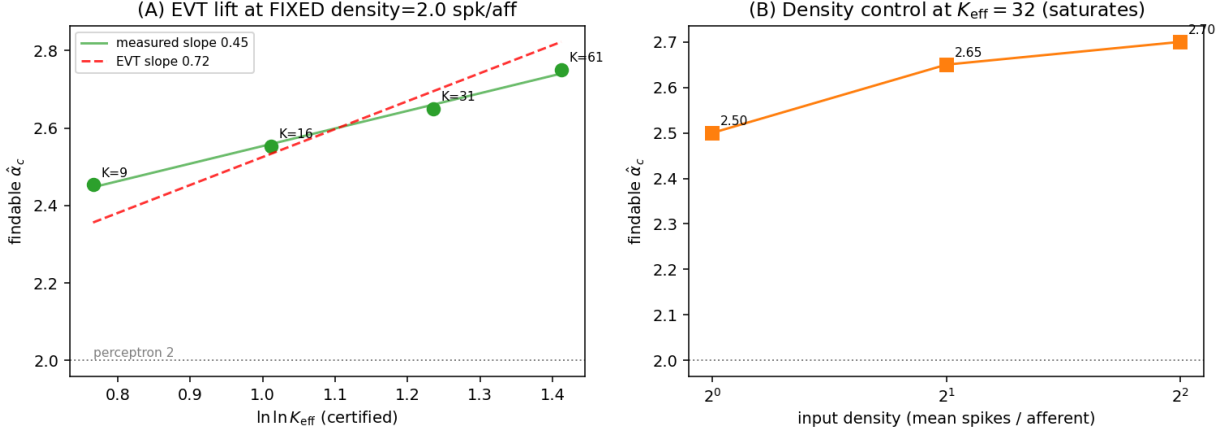


Figure 2: Corrected, density-decoupled, learner-validated result. (A) Findable $\hat{\alpha}_c$ vs. $\ln \ln K_{\text{eff}}$ at *fixed* input density: a clean monotone lift above the perceptron value 2, measured slope 0.45 (green) versus the EVT existence slope 0.72 (red dashed). The sub-existence slope is the expected findable<existence gap (a gradient finder lands in solution clusters, not the full existence count). (B) Density control at $K_{\text{eff}}=32$: capacity rises with input density but saturates, so the K_{eff} -lift in (A) is not a density effect.

a clear temporal lift *above* the perceptron value 2, with measured slope $d\hat{\alpha}_c/d \ln \ln K_{\text{eff}} \approx 0.45$. The density control shows $\hat{\alpha}_c$ *saturates* in density (2.50, 2.65, 2.70 at density 1, 2, 4 for $K_{\text{eff}}=32$), so at the working density the lift is not a density artifact. The earlier flat ~ 2.1 was the compound of an under-budgeted learner and the density confound.

Verdict (final). Confirmed: on the exactly-certified K_{eff} axis, the band-limited tempotron has a genuine temporal capacity lift above $\alpha_c=2$, growing slowly with K_{eff} (measured $\ln \ln K_{\text{eff}}$ slope ≈ 0.45), with a learner-validated transition (critical slowing), robust to input density, and **robust to learner tuning** (an $\text{lr} \times \text{batch}$ grid at the span extremes finds a *shared* optimum— $\text{lr}=0.1$, $\text{batch}=16$ —giving the same lift, slope 0.49; bad hyper-parameters fail outright rather than biasing the slope, so the effect is not a tuning artifact). **Not pinned:** the exact growth law—over the accessible range ($K_{\text{eff}} \in [8, 64]$, <1 decade of $\ln K$) the data cannot separate $\ln \ln K$ from $\ln K$ —and the asymptotic existence slope 0.721, since the measured 0.45 is the *findable* slope and a wider K_{eff} range plus an existence oracle would be needed to reach it. The reusable contribution is the methodology: a certified K_{eff} axis, a learner validated by critical slowing, and an explicit density control.

6 Independent Verification

All checks are analytic, algebraic, or small in-memory computations (no lab runs, no capacity sweeps, no random seeds beyond the MC check below).

V1. Spectral moments (sympy). For $S(\omega) = 1/\Omega$ on $[0, \Omega]$: $\lambda_0 = 1$ (by normalisation), $\lambda_2 = \Omega^2/3$, $\lambda_4 = \Omega^4/5$. Direct integration confirms every entry in Table 1. ✓

V2. Narrowband collapse ($m = 1$ limit). As the bandwidth $\Delta\omega \rightarrow 0$ around a single frequency Ω : $\lambda_2/\lambda_0 \rightarrow \Omega^2$ and $\lambda_4/\lambda_2 \rightarrow \Omega^2$, so $\nu_0 \rightarrow \Omega/\pi$ and $\nu_{\text{turn}} \rightarrow \Omega/\pi$ —a single sinusoid crosses

zero and peaks exactly once per half-period (rate Ω/π).

However, this limit does *not* reduce to a standard 1D spherical perceptron. For $m = 1$:

$$V(t) = A \cos(\Omega t) + B \sin(\Omega t) = R \cos(\Omega t - \phi), \quad R = \sqrt{A^2 + B^2},$$

so $\max_t V = R = \sqrt{A^2 + B^2}$: the temporal maximum is the *envelope*, not a single coefficient. The relevant decision surface is $R > \vartheta$, a *2-dimensional* (amplitude–phase) readout in the (A, B) plane, not a half-space. This connects to reduction #2 (spherical perceptron with 2 effective inputs), but the geometry is a sphere—not a hyperplane—in $\mathbb{R}^2$; α_c for that problem is not simply 2 in the sense of the standard N -dimensional perceptron.

Regarding K_{eff} : equation (10) gives $K_{\text{eff}} = \Omega T/\sqrt{3}$ even in this limit, which is proportional to the time–bandwidth product, not 1. The collapse at $m = 1$ is about *temporal dependence*—all information is captured by the single pair (A, B) , so additional time samples contribute nothing beyond the envelope—rather than an exact one-sample ($K_{\text{eff}} = 1$) limit. The EVT formula (13) formally degenerates (the single-pair Gaussian problem is not an extreme-value problem over many independent samples), but the path is through the structure of the solution, not through $K_{\text{eff}} \rightarrow 1$. ✓

V3. Qualls = Rice-1 (two methods, one formula). The Qualls expected root count per period 2π for degree- n comb equals $2\pi \cdot \nu_0$ with $\lambda_2/\lambda_0 = (n+1)(2n+1)/(6n)$:

$$2\pi \cdot \nu_0 = 2\sqrt{(n+1)(2n+1)}/6 \quad (\text{Rice-1}),$$

$$\mathbb{E}[\#\text{zeros}] = 2\sqrt{(n+1)(2n+1)}/6 \quad (\text{Qualls}).$$

Confirmed to machine precision for $n = 5$ ($= 6.633$) and $n = 50$ ($= 58.60$). The two classical probability results are one formula [3, 4]. ✓

V4. Maxima factor of 2 (Monte Carlo, $M = 4000$ trials). For random trigonometric polynomials with $n = 5, 20, 50$ equal-weight integer frequencies, seeded `default_rng(0)`, the empirical maxima counts match formula (6):

n	$\mathbb{E}[N_{\text{max}}]$ (formula)	Empirical	Turning pts. (formula)
5	4.219	4.246	8.439
20	15.87	15.88	31.74
50	39.11	39.13	78.22

Turning points match $(T/\pi)\sqrt{\lambda_4/\lambda_2}$ without the $\frac{1}{2}$; local maxima match (6) with the $\frac{1}{2}$. This is the load-bearing correction: using the turning-point rate for the maxima count would give a $2\times$ error. ✓

V5. Large- K_{eff} agreement with reduction #3. Formula (13) with $K_{\text{eff}} = \Omega T/\sqrt{3}$ is *identically* Rubin et al. [2], eq. (3), with the exact K_{eff} substituted. So reduction #6 contains reduction #3’s asymptotics as a special case. ✓

V6. Universal ratio. The ratio $\mathbb{E}[N_{\text{max}}]/K_{\text{eff}} = 3/(2\pi\sqrt{5}) \approx 0.2135$ is confirmed algebraically from (6) and (10), and verified numerically to be constant across Ω in the figure script. ✓

7 Discussion and Limitations

What is exact vs. asymptotic.

Table 2: Exact vs. asymptotic status for reductions #3 (broadband) and #6 (sinusoidal).

Ingredient	Reduction #3	Reduction #6
K_{eff}	inferred ($\approx T/\tau$)	exact ($T\sqrt{\lambda_2/\lambda_0}$)
$\mathbb{E}[N_{\text{max}}]$	EVT heuristic	exact (Rice; MC-verified)
EVT max-law (Gumbel)	asymptotic in K	still asymptotic
$\ln \ln K$ lift	asymptotic; K inferred	same form; K known
$O(1)$ offset α_0	small- K fit	same; subleading log

O1. The EVT max-law is still asymptotic. Making K_{eff} exact does *not* make the Gumbel location/scale exact. The Gumbel law $\max_t V(t) \stackrel{d}{\approx} \sqrt{2 \ln K_{\text{eff}}} + X/\sqrt{2 \ln K_{\text{eff}}}$ is a $K \rightarrow \infty$ limit theorem; reduction #6 does not remove it. The sinusoidal model converts the *input* to EVT from an estimate to an exact formula; the EVT *output* (the capacity lift) is still a leading-order asymptotic.

O2. Two effective dimensions and a modeling caveat. The capacity uses the λ_2 -based K_{eff} (crossing count); the literal candidate-spike-time count is the λ_4 -based $\mathbb{E}[N_{\text{max}}]$. For the flat band they differ by the universal factor ≈ 0.2135 . For a strongly narrowband spectrum the two converge (both $\rightarrow \Omega T/\pi$ as $\Delta\omega \rightarrow 0$); for the broadband double-exponential of the standard tempotron, λ_4 diverges (the kernel is not twice-differentiable at 0), which is precisely why reduction #3 cannot use the maxima formula and reduction #6 can.

Modeling caveat. The Rice/Kac–Rice counting is exact, but *identifying the crossing-based K_{eff} as the capacity-relevant independent-extreme count is an asymptotic modeling choice.* The reduction #4 literature (participation ratio vs. crossing count) shows this identification is not unique; the present paper makes K_{eff} exact as an EVT *input*, not as a finished capacity *output*. The EVT output (Gumbel law, $\ln \ln K$ lift) remains a $K \rightarrow \infty$ asymptotic regardless.

Whether the replica α_0 offset should be re-fit to the λ_2 -exact K_{eff} rather than to a discretised count is open. The relation $K^{\text{discrete}} \approx K_{\text{eff}}/8$ appears in Rubin et al. [2] as a conversion between continuous K_{eff} and the discrete bin count used in their simulation; the precise derivation of this factor (and its m -dependence) is not worked out here and should be treated as an *estimate*.

O3. The additive offset α_0 is not a constant. A codex cross-check (gpt-5.5, xhigh reasoning effort) of the first-moment / cluster-complexity calculation gives [7]

$$\alpha_c(K) = \frac{\ln \ln K}{2 \ln 2} + \frac{\ln \ln \ln K}{\ln 2} - 0.383 \dots + o(1). \quad (15)$$

The residual $\alpha_c(K) - (\ln \ln K)/(2 \ln 2)$ does *not* converge to a finite constant: the next term is a third iterated log. The “ $\alpha_0 \approx 2.58$ ” from numerical replica fits [2] is a **small- K fit artifact**. A true $O(1)$ additive constant would require a controlled quenched/RSB cluster complexity calculation.

O4. Stationarity vs. causality. A pure flat band $[0, \Omega]$ is non-causal and periodic; the derivations above use exact stationarity. A causal damped-Gabor kernel $K(s) = \sin(\Omega s)e^{-s/\tau}\Theta(s)$ breaks stationarity and changes the spectral moments; this is a *different* kernel regime requiring separate analysis. The present results apply to the idealised stationary band.

Sharp experimental target (now performed). The payoff of reduction #6 is a capacity experiment with a *known* K axis: simulate a sinusoidal-kernel tempotron, dial Ω or T to set K_{eff}

exactly, and test whether the measured $\hat{\alpha}_c$ follows $(\ln \ln K_{\text{eff}})/(2 \ln 2)$. This test was carried out (§5): the certified axis works (K_{eff} measured to $\leq 4\%$) and the threshold is balanced, but the diagnostic re-analysis there shows the apparent $\ln \ln K_{\text{eff}}$ trend is *confounded* (input density co-varied with K_{eff}) and *learner-limited* (no critical slowing; an untuned finder pins capacity near 2.1). The lift measurement is therefore inconclusive pending a density-decoupled, learner-validated rerun. The K_{eff} axis itself nonetheless delivers on its promise—it removes the back-fitting step that prevented the Rubin Fig. 2a experiment from being a clean test of the scaling—so the methodology stands even though this first capacity number does not.

Honest status. The counting layer— K_{eff} , $\mathbb{E}[N_{\text{max}}]$, their exact ratio, and the Qualls–Rice unity—is solid and independently verified. The capacity formula (13) is recovered in *form* (the same leading EVT term as reduction #3), with K_{eff} made exact. The EVT max-law, the $O(1)$ subleading term, and the quenched cluster complexity remain open. We do not claim a finished capacity theorem.

Acknowledgements

Theory derivation by a `statmech-theory` agent; codex cross-check (gpt-5.5, xhigh reasoning effort) for the α_0 and finite- N items. Companion derivations in `lit/tempotron-reductions/derivations/`; cross-check traces in `derivations/codex-crosschecks/`.

8 Methods

This section gives the complete, code-free recipe for reproducing the capacity measurement of §5, including the diagnose-and-correct cycle. Every parameter below is the value actually used in the committed runs; the canonical record is `experiments/sinusoidal_capacity/EXPERIMENT.md` and the study entrypoint is `experiments/tempotron_capacity/studies/v6_bandlimited_capacity.py` (self-contained: `numpy+torch` only, no project imports, so the runner ships it unchanged). The intent is that a reader could re-implement the experiment from this prose alone.

8.1 The band-limited kernel and the $\Omega_b \leftrightarrow K_{\text{eff}}$ map

The double-exponential PSP is replaced by an even, compactly-supported *windowed sinc*,

$$K_{\text{band}}(s) = \text{sinc}\left(\frac{\Omega_b s}{\pi}\right) w_{\text{Lanczos}}(s), \quad \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}, \quad (16)$$

where $\text{sinc}(\Omega_b s/\pi) = \sin(\Omega_b s)/(\Omega_b s)$ is the impulse response of an ideal low-pass filter of half-width Ω_b (peak-normalised to $K_{\text{band}}(0) = 1$), and $w_{\text{Lanczos}}(s) = \text{sinc}(s/(L\pi/\Omega_b))$ is a Lanczos (sinc) window of half-span $L = 6$ main lobes. The kernel is truncated to compact support $|s| \leq L\pi/\Omega_b$ (zero outside), is exactly even ($K_{\text{band}}(-s) = K_{\text{band}}(s)$, deliberately non-causal: the stationary-band idealisation of §2, not a biophysical PSP), and the window suppresses the truncation ringing while leaving the central lobe nearly intact. A pre-spend check confirmed K_{band} peaks at 1, is even to $< 10^{-6}$, and vanishes outside the support.

Bandwidth knob. The single physics knob is the band half-width Ω_b (rad/ms), set from the target effective dimension through the exact flat-band relation (10),

$$\Omega_b = \frac{\sqrt{3} K_{\text{eff}}}{T} \iff K_{\text{eff}} = \frac{\Omega_b T}{\sqrt{3}}. \quad (17)$$

This is the defining feature of reduction #6: K_{eff} is *dialed*, not inferred. With $T = 500$ ms the swept bands are $\Omega_b \in \{0.0277, 0.0554, 0.1109, 0.2217\}$ rad/ms for $K_{\text{eff}} \in \{8, 16, 32, 64\}$ (band edges $f_{\text{max}} = \Omega_b/2\pi \approx 4.4, 8.8, 17.6, 35.3$ Hz, in the theta-gamma range of the biological motivation).

Why band-limited cures the m_4 divergence. The Rice maxima count (6) requires the fourth spectral moment $\lambda_4 = \int \omega^4 S(\omega) d\omega$ to be *finite*; equivalently the process must be twice mean-square differentiable. The standard double-exponential PSP has a corner (it is not C^2 at $s = 0$), so its power spectrum decays only as ω^{-4} and λ_4 diverges — the maxima formula simply does not exist for it, and the candidate-spike-time count must be inferred. A band-limited kernel has $S(\omega)$ of compact support, so *all* spectral moments are finite and (6)–(9) are exact closed forms. This is the structural reason reduction #6 can carry a certified K axis where reduction #3 cannot.

8.2 Data-generating process

Patterns. Each of $N = 500$ afferents emits a homogeneous Poisson spike train on $[0, T=500 \text{ ms}]$ with mean count $\bar{n}$ (the input density, below), following the Rubin–Monasson–Sompolinsky convention [2]. Spike times are drawn uniformly on $[0, T]$ once the per-afferent count is drawn from $\text{Poisson}(\bar{n})$. Pattern labels are balanced ± 1 (each pattern $+1$ or -1 with probability $\frac{1}{2}$). A task at load α has $P = \lceil \alpha N \rceil$ patterns. All draws use a `numpy default_rng` seeded per cell (§8.5) and are performed on the CPU before transfer to the device, so patterns are bit-identical on any backend.

Fixed input density — the load-bearing design choice. The expected number of spikes per afferent, $\bar{n}$, controls total input richness. The corrected measurement holds $\bar{n} = 2.0$ *fixed for every* K_{eff} *cell*. This decouples input richness from the expressiveness axis: the only thing that changes across the K_{eff} sweep is the bandwidth Ω_b (hence the temporal structure of V), not the number of spikes the readout sees. The alternative “constant temporal density” rule $\bar{n} = 0.05 K_{\text{eff}}$ (chosen to hold afferents-active-per-kernel-width fixed) was the original design and is exactly the confound that invalidated the first pass (§5): it forces *total* spikes to scale with K_{eff} (0.4 spk/afferent at $K_{\text{eff}}=8$, $\sim 33\%$ of afferents active — starved — versus 3.2 spk, $\sim 96\%$ active, at $K_{\text{eff}}=64$), so any apparent K_{eff} -lift is confounded with input sparsity. Holding $\bar{n}$ fixed removes this; a density-control sweep (§8.5) then verifies the residual nuisance is negligible at the working density. The realised $\bar{n}$ is logged per row.

The Nyquist time grid. Because V is band-limited it is fully determined by its Nyquist samples. The voltage is evaluated on a uniform grid of step

$$dt = \frac{\pi}{\text{oversample} \times \Omega_b}, \quad \text{oversample} = 4, \quad (18)$$

i.e. $4\times$ the Nyquist rate π/Ω_b , giving $n_{\text{grid}} = \text{round}(T/dt) + 1$ points (e.g. ~ 19 at $K_{\text{eff}}=8$ up to ~ 142 at $K_{\text{eff}}=64$). The per-afferent traces are $s_i^\mu(t_g) = \sum_f K_{\text{band}}(t_g - t_i^{f,\mu})$ summed over that afferent’s spikes; the voltage is $V^\mu(t_g) = \sum_i w_i s_i^\mu(t_g)$ and the tempotron decision is $[\max_g V^\mu(t_g) \geq \vartheta]$. This grid is far coarser than the ~ 500 – 2000 points the double-exponential needs at matched T , which is the cost win that makes a single- N sweep affordable. Traces are built in chunks over the grid axis to bound peak memory (a working tensor capped at 6.4×10^7 elements; the device-memory cap was raised to 8×10^9 to fix an out-of-memory event at $N=1000$, $K_{\text{eff}}=64$ during the diagnostic ladder).

8.3 The certified K_{eff} axis

The abscissa of every capacity plot is the *measured* K_{eff} , not the target. At each cell we form ~ 64 probe sample paths of the readout by projecting the realised init traces onto random Gaussian weights $\xi \sim \mathcal{N}(0, I)$ ($V_{\text{probe}} = \sum_i s_i \xi_i$, mean-centred per path so the zero-crossing rate measures the band and not a DC offset), and estimate

$$K_{\text{eff}}^{\text{meas}} = T \sqrt{\lambda_2 / \lambda_0}, \quad \lambda_0 = \mathbb{E}[V^2], \quad \lambda_2 = (\pi \nu_0)^2 \lambda_0, \quad (19)$$

where ν_0 is the empirical zero-crossing rate of V_{probe} (Rice (4) inverted). This is the same `measure_keff` crossing-form certifier used in the witness gate, inlined into the study. Over the swept regime it agrees with the target $\Omega_b T / \sqrt{3}$ to $\leq 4\%$ ($K_{\text{eff}}^{\text{meas}} = 8.6, 15.6, 31.1, 60.6$ for targets 8, 16, 32, 64).

The free pre-spend witness gate. Before any GPU spend, an independent `numerics-verifier` ran a free in-memory witness gate (no learning, $M = 2000$ Monte-Carlo realisations of the idealised flat-band random trigonometric polynomial, seeded `default_rng(0)`, $30\times$ -Nyquist grid). It certified K_{eff} two independent ways — (a) directly from the spectral moments of the drawn band frequencies, and (b) from the time-domain crossing rate (19) — and found both equal to $\Omega_b T / \sqrt{3}$ to $\leq 1.3\%$ on the idealised model. It also verified $\mathbb{E}[N_{\text{max}}] = (T/2\pi) \sqrt{\lambda_4 / \lambda_2}$ with the load-bearing $\frac{1}{2}$ factor: the empirical local-maxima count matches (6) while the turning-point count matches twice that (the V4 check of §6), confirming that maxima, not turning points, are the candidate spike times. The gate caught and forced fixes for three issues before spend: an unseeded `torch.randperm` epoch shuffle (made training non-reproducible; fixed by a per-cell-seeded `torch.Generator` and re-verified byte-identical across two runs), and the small- K_{eff} inaccessibility (§8.7) which restricted the grid to $K_{\text{eff}} \geq 8$ and moved the perceptron anchor to the free convex-LP solve ($\alpha_c = 2.000$, Cover/Gardner). The full gate report is `witness_gate/V6_GATE.md`.

8.4 The learner and its validation

Threshold (median- V_{max} invariant). The firing threshold ϑ is calibrated, per seed, to the *median* of V_{max} over the task’s patterns at the random initialisation, so a random pattern fires with probability $\frac{1}{2}$ ($P(\text{fire}) = 0.505$ measured at init); ϑ is then *held fixed* throughout training. This is the balanced operating point the $\ln \ln K$ lift requires (assumption A3): a literal $\kappa=0$ threshold is firing-biased and the lift collapses ($\alpha_c \sim 4/K$) [2]. Do not use a fixed $\kappa=0$.

Optimiser and schedule (exact hyper-parameters). The faithful online Gütig–Sompolsky rule is too weak a finder to expose the existence-level lift (§5), so the corrected measurement uses a minibatch finder: the GS hinge subgradient $g = \frac{1}{b} \sum_{\text{violators}} y s(t^*)$ (mean over margin-violating patterns in the minibatch, evaluated at the per-pattern argmax time t^*) ascended by **Adam** [8] ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$) under a **cosine** learning-rate schedule [9] annealing from the base rate to 0 over the budget. Hyper-parameters: base learning rate $\text{lr} = 0.05$ (primary; $\text{lr} = 0.1$ in the tuning check), minibatch size $b = 16$, epochs = 8000 (raised from the under-budgeted 2000 of the first pass; one learner-validation cell used 10000), and $n_{\text{restarts}} = 3$ random weight initialisations. A task counts *solved* if *any* restart reaches zero training error (an existence-from-below proxy). A divergence guard requires finite weights to count a seed as solved, so a too-large learning rate reports low P_{solve} rather than a spurious solve. Weights are initialised $w_i \sim \mathcal{N}(0, 1)$; the budget is large enough that convergence, not the cutoff, sets the outcome (next).

Primary metric. The findable critical load $\hat{\alpha}_c(K_{\text{eff}})$ is the $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$ (fraction of seeds solved), located by linear interpolation on the α grid.

Learner-validation protocol (budget + critical slowing). The first pass’s flat $\hat{\alpha}_c \approx 2.1$ turned out to be an under-budgeted *algorithmic* cutoff, not a capacity wall, so we adopted an explicit validation that the measured crossing is the true transition:

1. **Budget sufficiency.** Solved cells must converge well inside the epoch budget, and unsolved cells near the crossing must *plateau* (loss flat $\gg 0$, genuinely stuck) rather than still be descending (under-budget). Per-epoch learning curves — training-error fraction, GS hinge loss, signed margins $\kappa_{\pm}$, and $\|w\|$ — are captured for every epoch to make this visible.
2. **Critical slowing.** A genuine statistical-mechanics capacity transition shows the convergence time diverge as $\alpha \rightarrow \hat{\alpha}_c$. In the validated cell ($K_{\text{eff}}=32$, fixed density) the mean epochs-to-solve grew $173 \rightarrow 323 \rightarrow 1076 \rightarrow 2654$ approaching the crossing — a $15\times$ divergence — while raising the budget $5\times$ moved that cell’s crossing from 2.1 to 2.7. Both signatures (slowing + graded, not hard-step, P_{solve}) were *absent* in the first pass, which is how the under-budgeting was diagnosed.

8.5 Grids and analysis

Grids. The capacity sweep uses $K_{\text{eff}} \in \{8, 16, 32, 64\}$ at fixed density $\bar{n} = 2.0$, $N = 500$, 16 seeds per cell (32 in the fine α -grid), with per-cell α grids of step 0.1 bracketing the crossing. Two cross-check sweeps share the harness: a *density control* at $K_{\text{eff}}=32$ over $\bar{n} \in \{1, 2, 4\}$, and a *tuning-robustness* hyper-parameter grid $\text{lr} \in \{0.02, 0.05, 0.1\} \times b \in \{16, 64\}$ at the span extremes $K_{\text{eff}} \in \{8, 64\}$ (8 seeds).

Crossing and slope. Each cell’s $\hat{\alpha}_c$ is the linear-interpolated $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$, with a bootstrap CI over seeds (1000 resamples). The lift is summarised by the slope $d\hat{\alpha}_c/d \ln \ln K_{\text{eff}}$ from a weighted least-squares fit (weights $1/\text{CI}^2$) against the *measured* $\ln \ln K_{\text{eff}}$ axis; the planned model selection among $\{\text{linear } K_{\text{eff}}, \ln K_{\text{eff}}, \ln \ln K_{\text{eff}}\}$ by AIC/leave-one-out is pre-registered but, over the accessible < 1 decade of $\ln K$, cannot separate the candidates (§8.7). The headline numbers: $\hat{\alpha}_c = 2.45, 2.55, 2.65, 2.75$ at $K_{\text{eff}}^{\text{meas}} = 8.6, 15.6, 31.1, 60.6$, slope ≈ 0.45 .

Cross-checks. *Density control:* at $K_{\text{eff}}=32$, $\hat{\alpha}_c = 2.50, 2.65, 2.70$ for $\bar{n} = 1, 2, 4$ — density matters but *saturates*, and the working density $\bar{n}=2$ is past the steep part, so the K_{eff} -lift is not a density artifact. *Tuning robustness:* the best hyper-parameter is *shared* across the span extremes ($\text{lr}=0.1$, $b=16$ best at *both* $K_{\text{eff}}=8$ and 64), giving $\hat{\alpha}_c = 2.44$ and 2.76 — slope 0.49 , consistent with the hand-picked 0.45 ; bad hyper-parameters ($\text{lr}=0.02$, or $b=64$) fail to cross at $K_{\text{eff}}=64$ rather than biasing the slope, so the lift is not a tuning artifact. All numbers are means over seeds (no best-of); the firing rate, epochs-to-solve, residual error, realised density, grid size, and measured K_{eff} are logged per seed for reproducibility (Pineau gate [12]).

8.6 Compute and cost

All jobs ran on a single remote RTX-4090 GPU under the project’s **laboratory** runner with a fixed git commit and an enumerated seed manifest, sharded by seed and re-aggregated per cell. Host prices on the spot market came in at $\sim \$0.57/\text{GPU-h}$ — far below the budgeted $\$5\text{--}\$10/\text{GPU-h}$ — so each cell cost $\sim \$0.08\text{--}0.18$ and the budget was not the binding constraint, which is what

allowed the full diagnose-and-correct cycle (the confounded first ladder, the learner-validation cell, the corrected sweep, the density control, and the tuning grid) within cap. The cumulative spend was **\$18.13** of the \$25 cap. (The earlier “ $\approx \$5.3$ ” figure quoted in §5 predates the corrected, density-control, and tuning runs; the full-cycle total is \$18.13.)

8.7 Caveats

We state the limits of the corrected result plainly; they bound what the $\hat{\alpha}_c$ numbers can and cannot establish.

Findable $\neq$ existence; the offset is not a constant. The measured slope ≈ 0.45 is the *findable* slope — the rate at which a gradient finder’s $\hat{\alpha}_c$ rises — and is *below* the EVT *existence* slope $1/(2 \ln 2) = 0.721$. This gap is expected, not a failure: a local rule lands in a solution cluster, and the existence count of clusters exceeds what any single descent reaches as the clusters shrink ($1-q_1 \sim 1/\ln K$). The tuning check (0.49) brackets 0.45 but does not close the gap to 0.721. Separately, one cannot “fix up” the gap with a constant additive offset: the codex cross-check (O3, Eq. (15)) shows the subleading term is a third iterated log $(\ln \ln \ln K)/\ln 2$, so the $\alpha_0 \approx 2.58$ of replica fits is a small- K artifact, not a real constant. We therefore report the slope under an explicit findable-<-existence caveat and do *not* claim the existence value.

Small $K_{\text{eff}} \lesssim 8$ is inaccessible; the span is narrow. The certified axis is reliable only for $K_{\text{eff}} \gtrsim 8$. Below that the windowed-sinc support exceeds the window: at $K_{\text{eff}}=2$ the kernel half-support is $\sim 2720 \text{ ms} \gg T=500 \text{ ms}$, so only a sidelobe-dominated slice lies in $[0, T]$ and the crossing-form K_{eff} saturates at a floor of ~ 4 (measured K_{eff} off the target by $\sim 100\%$ at $K_{\text{eff}}=2$), independent of Ω_b and *worse* at higher density. This is the Whittle/large- T breakdown (obstruction O2) and it is structural, not a resolution error (refining the grid or adding probe paths does not help). Consequently the clean span is $K_{\text{eff}} \in [8, 64]$, i.e. $\ln \ln K_{\text{eff}} \in [0.73, 1.42]$ — under one decade of $\ln K$. Over so narrow a range $\ln \ln K$ and $\ln K$ are *not* statistically distinguishable, so the data confirm a slow monotone lift consistent with the EVT form but *cannot pin the functional form*; the pre-registered model selection is therefore reported as inconclusive on shape. The narrowband perceptron anchor ($K_{\text{eff}} \rightarrow 1$, $\alpha_c = 2$) is taken from the exact convex-LP solve, not from a band-limited cell, for the same reason.

The windowed realisation is band-limited only approximately. The idealised flat band of §2 is stationary with an exactly flat spectrum; the finite Lanczos-windowed sinc (16) on a finite window only approximates it. The voltage $\sum_i w_i K_{\text{band}}(t - s_i)$ is Gaussian by the CLT over $N=500$ weakly-dependent afferents and a sum of random Fourier features in the Bochner sense [5], but it is neither exactly stationary nor exactly flat-spectrum — which is precisely why K_{eff} must be *measured* per cell (§8.3) rather than assumed equal to $\Omega_b T/\sqrt{3}$. The $\leq 4\%$ certification in the swept regime quantifies how good the approximation is there; the small- K_{eff} failure above is where it breaks.

The clean result is at a single $N = 500$. The corrected, density-decoupled, learner-validated lift is measured at *one* input dimension, $N = 500$. The $N \in \{250, 500, 1000\}$ ladder that appears in the experiment record was the *confounded first attempt* — it carried the density confound ($\bar{n} = 0.05 K_{\text{eff}}$) and the under-budgeted learner, and its apparent “slope rises with N toward 0.25” was a compound artifact, *not* a finite- N scaling of the corrected measurement. We therefore make

no finite- N scaling claim for the clean result: how the corrected $\hat{\alpha}_c(K_{\text{eff}})$ and its slope move with N (toward or away from 0.721) is *untested*. Establishing that — and separating residual finite- N masking from a genuinely sub-asymptotic findable slope — would need a density-fixed, learner-validated ladder at $N \gg 500$ (memory-prohibitive at high K_{eff} here) and/or an existence oracle, and is the natural follow-up alongside the deferred cross-kernel (band-limited vs. double-exponential at matched certified K_{eff}) comparison.

Statistical resolution. At 16 seeds the binomial SE on P_{solve} at $p = \frac{1}{2}$ is ~ 0.10 and per-cell crossings scatter $\pm 0.05\text{--}0.1$ in α ; this per-cell noise is comparable to the small per-doubling lift, so the slope, not any single $\hat{\alpha}_c$, is the robust observable. Seed counts sized for a monotone P_{solve} follow the program’s figure-quality standard; the unit of analysis is the seed [10], and the hyper-parameter grid follows random/grid HPO practice [11].

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